

Organic & Supramolecular Chemistry

Aerobic α -Sulfonylation of Ketones with Zinc Sulfinates in Aqueous Micellar Media: Highly Selective Access to β -Keto Sulfones

Ying Fu,* Quan-Zhou Li, Ming-Peng Li, and Zhengyin Du*[a]

An efficient, inexpensive and environmentally friendly synthesis of β -ketosulfones, via NaI-catalyzed oxidative sulfonylation of ketones with zinc sulfinates, employing ethylene dibromide (EDB) and air as the oxidants, is described. EDB was shown to be a mild organic oxidant to convert NaI into molecular iodine that promote the cross-coupling reactions of zinc sulfinates with ketones, producing β -ketosulfones. Mechanistic studies indicated that a radical pathway might be involved in the reaction process. The key feature of the present protocol is the far greater solubility of NaI and sulfinates in micellar arrays present in aqueous solution where the oxidative coupling reactions are taking place.

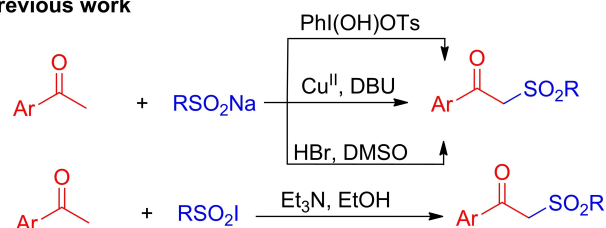
Organic reactions are commonly carried out in organic solvents which, being derived from petroleum, are volatile, flammable and toxic. Thus reactions proceeded in water solution represent the most environment-friendly methods that decreasing the huge amounts of organic waste generated by use of organic solvents in organic syntheses.^[1] However, water itself cannot replace organic solvents because many organic chemicals do not dissolve and, of course, do not react in water. TPGS-750-M (DL- α -Tocopherol methoxypolyethylene glycol succinate)^[2] is a designer surfactant that forming micelles upon dissolution in water spontaneously. Thus, reactants and catalysts are coexisted in high concentrations within the lipophilic cores of micellar arrays, making reactions to take place in water medium.

The β -ketosulfone containing compounds have been extensively studied because of their widespread presence in pharmaceuticals, agrochemicals and biologically active compounds.^[3] In addition, β -ketosulfones are key building blocks for the construction of many biologically active heterocycles, e.g. quinolines,^[4] γ -sulfonyl δ -lactones,^[5] oxabenz[3.3.1]bicycles,^[6] dihydrofurans,^[7] polyfunctionalized 4H-pyrans,^[8] lycopodium^[9] and pyrazolines^[10] etc. As a consequence, considerable efforts have been devoted to developing efficient

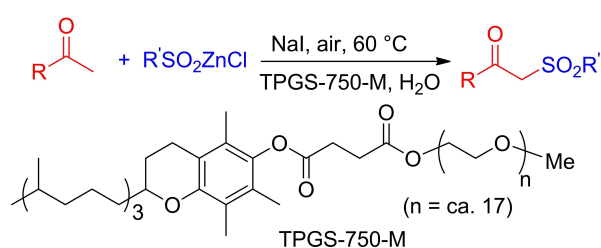
methods for the preparation of this framework. Commonly, β -ketosulfones are prepared by the nucleophilic substitution reaction of toxic α -halogenated aryl ketones with sodium sulfinates.^[11] However, this protocol encompasses some obvious limitations such as the prefunctionalized materials, relatively complicated or harsh reaction conditions and undesired by-products. Alternatively, procedures such as oxidation of β -ketosulfides,^[12] sulfonylation of alkenes,^[13] alkynes^[14] and enol acetates^[15] etc. have been successively developed to access structurally diverse β -ketosulfone derivatives, employing sulfinic acids, sulfonylhydrazides, arenesulfinate salts, sulfonyl chlorides or thiophenols as the sulfonyl candidates. Nevertheless, limitation for practical syntheses remains as the starting materials of above mentioned protocols are commonly more expensive than β -ketosulfone products.

From a green and sustainable chemistry perspective, the direct oxidative coupling of ketones with sulfinate salts, both of which are easily accessible chemicals, is therefore an ideal method for the construction of β -ketosulfones. Early in 2006, Kumar and colleagues^[16] (Scheme 1) reported the first direct syntheses of β -ketosulfones from ketones through hypervalent iodine reagents mediated nucleophilic substitution of the in situ generated α -tosyloxyketones with sodium arenesulfi-

Previous work



This work

Scheme 1. Methods for β -ketosulfones syntheses from ketones.

[a] Prof. Dr. Y. Fu, Q.-Z. Li, M.-P. Li, Prof. Dr. Z. Du

College of Chemistry and Chemical Engineering, Northwest Normal University, 967 Anning East Road, Lanzhou, P. R. China
E-mail: fu_yingmail@126.com

zhengyin_du@nwnu.edu.cn

 Supporting information for this article is available on the WWW under <https://doi.org/10.1002/slct.202000539>

nates. Later in 2014, Sreedhar et al. disclosed a convenient procedure for syntheses of β -keto sulfones via nucleophilic addition of the in situ generated enolates to sulfonyl iodides.^[17] Shortly afterwards, a mild CuBr_2/DBU catalyzed cross-coupling reactions between aryl ketones and sodium sulfinates at room temperature was reported.^[18] Most recently, Tang group described a convenient oxidative sulfonylation of aryl ketones employing sodium sulfinates as the sulfone source and DMSO as the oxidant.^[19] Although these procedures are mild and efficient, they all have limits in practical laboratory synthesis because of the employment of expensive hypervalent iodine reagents or limited starting materials (some protocols are only effective to aryl ketones), thus a general and efficient method to the synthesis β -ketosulfones is therefor highly desired.

During the past decade, we have interests on the synthesis of sulfur containing compounds employing sulfonyl chlorides as the sulfur candidates.^[20] Previously, we reported a reductive sulfonylation of alkyl chlorides with sulfonyl chlorides whereby various type of sulfones were prepared in good to high yields.^[20d] Recently, We reported a convenient NaI -catalyze oxidative amination of aromatic sodium sulfinates employing both air and ethylene dibromide (EDB) as the oxidants.^[20c] Our previous findings led us to investigate the more challenging reaction of sulfonylation of ketones with chlorozinc sulfinates,^[21] which could be easily obtained from corresponding sulfonyl chlorides in nearly quantitative yields in our laboratory,^[20d] for the synthesis of β -ketosulfones.^[22]

Our initial trial with $p\text{-TolSO}_2\text{ZnCl}$ (**1a**, 1.5 mmol) and acetophenone (**2a**, 1.0 mmol) in the presence of two equivalents of EDB, NaHCO_3 (2.0 mmol) in DMF at 60°C for 8 h did not initiate the desired reaction (Table 1, entry 1). To our delight, when 2 equivalents of Na_2CO_3 was introduced as the base, the coupling reaction occurred at 60°C in DMF and the desired product **3aa** was obtained in 33% yield (entry 2). Replacement of Na_2CO_3 with K_2CO_3 as a base further improved the yield of **3aa** to 42% (entry 3). Interestingly, when water alone was introduced as the solvent, the yield of **3aa** was further improved to 62% yield (entry 4). However the use of

organic bases such as Et_3N , DIPEA (N,N-Diisopropyl ethylamine) and DBU (1,8-Diazabicyclo[5.4.0]undec-7-ene) did not improve the yield of **3aa** further (entries 5–7). The amount of EDB plays a key role in the oxidative coupling of acetophenone and chlorozinc sulfinates as was denoted in entry 8, when only one equivalent of EDB was employed, the yield of **3aa** was dramatically decreased to 36% yield. The best yield of **3aa** was obtained when 2 wt.% of TPGS-750-M was introduced (Table 1, entry 9). A control experiment proceeded under nitrogen atmosphere showed that oxygen in air might play a role in this transformation also (entry 10).

With the optimized reaction conditions in hand (table 1, entry 9), the scope of ketones for this reaction was then examined with $p\text{-TolSO}_2\text{ZnCl}$ (**1a**) (Table 2). First, acetophenone bearing a strong electron-withdrawing group (i.e. NO_2) proceeded this reaction quite well and yielded **3ab** in 88% yield. Nevertheless, reactions of acetophenones with a halogen substituent at para position always give a slightly lower yields of products (**3ac–3ae**). Acetophenones bearing an electron donating group (i.e. Me, OMe) generally produce good yields of β -ketosulfone products (**3af & 3ag**). Other aryl alkyl ketones screened such as 1-acetonaphthone, 1-indanone and 2-acetothienone are all capable to perform this transformation and deliver the corresponding products in 72% to 79% yields (**3ah–3ak**). Notably, dialkyl ketones, as exemplified by acetone, cycloheptanone and 1-cyclopropylethanone, are applicable materials in our reaction system, affording the desired products in good to high yields (**3al–3an**). Unexpectedly, the highly steric hindered aliphatic ketones, i.e. bicyclo[2.2.2]octane,

$p\text{-TolSO}_2\text{ZnCl}$ (1a) + PhCOCH_3 (2a) conditions $p\text{-TolSO}_2\text{CH}_2\text{C(=O)Ph}$ (3aa)			
Entry	Additive (mmol)	Solvent	Yield (%) ^[b]
1	NaHCO_3 (2.0)	DMF	0
2	Na_2CO_3 (2.0)	DMF	33
3	K_2CO_3 (2.0)	DMF	42
4	K_2CO_3 (2.0)	H_2O	62
5	Et_3N (2.0)	H_2O	41
6	DIPEA (2.0)	H_2O	58
7	DBU (2.0)	H_2O	55
8	K_2CO_3 (2.0)	H_2O	36 ^[c]
9	K_2CO_3 (2.0)	H_2O	87 ^[d]
10	K_2CO_3 (2.0)	H_2O	63 ^[e]

[a] **1a** (1.5 mmol), **2a** (1.0 mmol), EDB (2.0 mmol), solvent (4 mL), air, 60°C , 8 h. [b] Isolated yields. [c] 1.0 mmol EDB was used. [d] 2 wt.% TPGS-750-M was used. [e] Reaction was carried out under N_2 atmosphere.

Table 2. Scope of ketones. ^[a,b]			
$p\text{-TolSO}_2\text{ZnCl}$ + $\text{R}-\text{C}(=\text{O})-\text{R}'$ 1a 2	TPGS-750-M, NaI, EDB K ₂ CO ₃ , H ₂ O, air, 60 °C		$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})-\text{R}'$ 3ab-3an
$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{X}$ 3ab : X = 4-NO ₂ 88% 3ac : X = 4-F 59% 3ad : X = 4-Br 65% 3ae : X = 4-I 51% 3af : X = 4-OMe 73% 3ag : X = 3-Me 78%	$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_{10}\text{H}_7$ 3ah : 77% $p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_8\text{H}_7$ 3ai : 78% $p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_9\text{H}_6\text{Cl}$ 3aj : 79%	$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{Me}$ 3ak : 72% $p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{Me}$ 3ap : 0%	$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{CN}$ 3aq : 0%
$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{Me}$ 3al : 80% $p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_7\text{H}_{13}$ 3am : 57% $p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_7\text{H}_{13}$ 3an : 68%	$p\text{-TolSO}_2\text{CH}_2\text{C}(=\text{O})\text{C}_6\text{H}_4\text{CN}$ 3aq : 0%		

[a] **1a** (1.5 mmol), **2** (1.0 mmol), EDB (2.0 mmol), TPGS-750-M (2 wt.%), K₂CO₃ (2.0 mmol), water (4 mL), 60 °C under air for 8 h. [b] Isolated yields.

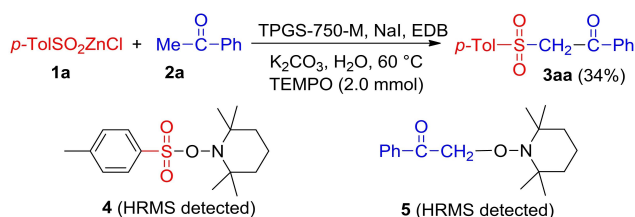
tropinone and 4,4-dimethyl-3-oxopentenenitrile do not participate this transformation at all (**3o–3q**).

Afterwards, we examined the scope of this protocol using various chlorozinc sulfonates. As shown in table 3, phenylsulfinyloxy zinc chloride reacted with 1-phenylpentan-1-one to give **3ba** in 54% yield. 4-Chlorophenylsulfinyloxy zinc chloride reacted with 1-indanone to give **3bb** in 77% yields. Benzene-sulfinyloxy zinc chlorides having CF₃ or plural halogen substituents (especially F) on the benzene ring worked similarly and produced the desired products in low isolated yields (less than 60%), possibly because these substituents (CF₃ and F) were partially hydrolyzed under this basic reaction condition, thus reduced the yields of isolated products. For examples, the reaction of 4-(trifluoromethyl)phenylsulfinyloxy zinc chloride with acetophenone in air afforded the corresponding β -ketosulfone (**3bc**) in 59% yield. Heteroaromatic sulfinate, as exemplified by quinolin-7-ylsulfinyloxy zinc chloride, could also be employed to react with 2'-methylacetophenone and produced **3bg** in a yield of 76%. Notably, aliphatic chlorozinc sulfonates are applicable substrates here. As exemplified by **3bh** to **3bj**, under our optimized conditions, normally good to high yields of corresponding β -ketosulfones could be obtained.

We next carried out control experiments to elucidate the reaction mechanism (Scheme 2). In the presence of two equivalents of TEMPO under our optimized conditions, the reaction between **1a** and **2a** gave only 34% yield of **3aa**. HRMS analysis on the reaction mixture showed that adducts **4**

$\text{RSO}_2\text{ZnCl} + \text{R}^1\text{C}(=\text{O})\text{CH}_2\text{R}^2 \xrightarrow[\text{K}_2\text{CO}_3, \text{H}_2\text{O}, \text{air}, 60^\circ\text{C}]{\text{TPGS-750-M, NaI, EDB}} \text{R}^1\text{C}(=\text{O})\text{CH}_2\text{C}(=\text{O})\text{R}^2 + \text{R}^1\text{SO}_2\text{CH}_2\text{R}^2$	
1	2
3ba 54%	3bb 77%
3bc 59%	3bd 46%
3be 43%	3bf 56%
3bg 76%	3bh 78%
3bi 65%	3bj 47%

[a] **1** (1.5 mmol), **2** (1.0 mmol), EDB (2.0 mmol), TPGS-750-M (2 wt.%), K₂CO₃ (2.0 mmol), water (4 mL), 60 °C, air, 8 h. [b] Isolated yields.

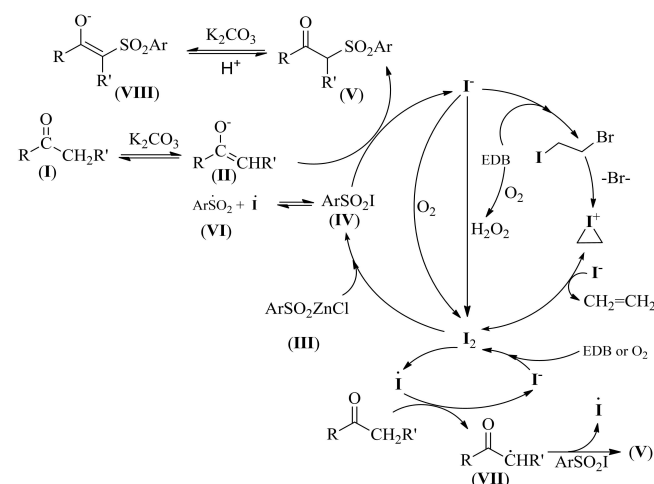


Scheme 2. control experiment.

and **5** were formed, demonstrating the occurrence of both acetophenone radical and sulfonyl radical, which indicated the operation of a single-electron-transfer pathway. Furthermore, these reactions are only effective under base condition, suggesting that the equilibrium amount of the enolate generated from a ketone in the presence of K₂CO₃ is most likely the active species that reacts with the sulfonyl iodide, in situ formed from chlorozinc sulfinate with iodine, to afford β -ketosulfone.

Based on these experimental results and our previous results,^[20c] we propose the catalytic cycles as shown in Scheme 3. The key steps include (i) EDB and oxygen oxidized iodide ions into elemental iodine which then converted chlorozinc sulfinate (**III**) into sulfonyl iodide (**IV**). (ii) Decomposition of **IV** provided the iodine radical and sulfonyl radical. In addition, **IV** could react with enolate (**II**) to give the β -keto sulfone product (**V**). (iii) Under heat, molecular iodine was activated into iodine radical which convert ketone into ketone radical (**VII**). (iv) Ketone radical was trapped with sulfonyl iodide (**IV**) to give the desired β -keto sulfone product (**V**) and regenerate the iodine radical. Notably, β -keto sulfone (**V**) encompasses a more acidic α -H than that of ketone (**I**) and thus, under basic condition, enolate (**VIII**) may be formed. However, enolate **VIII** is less nucleophilic than enolate **II** owing to the weaker basicity and higher steric constraints around the α -carbon, thus the second sulfonylating reaction on enolate (**VIII**) was suppressed.

In conclusion, we have described a mild, practical, and environmentally friendly NaI catalyzed oxidative coupling of chlorozinc sulfonates with ketones method to prepare β -ketosulfones. A wide range of functional groups can be tolerated in the reaction conditions and the corresponding β -ketosulfones were obtained in moderate to good yields. The key feature of this process is the greater solubility of NaI and sulfonates in micellar arrays where the oxidation is taking place.



Scheme 3. Proposed reaction mechanism.

Supporting Information Summary

Experimental details can be found in the Supporting Information.

Acknowledgements

The authors are grateful for financial support from the National Natural Science Foundation of China (No. 21762040, 21762039 and 21262030).

Conflict of Interest

The authors declare no conflict of interest.

Keywords: cross-coupling • ketones • radical reactions • synthetic methods • zinc

- [1] a) C.-J. Li, *Chem. Rev.* **2005**, *105*, 3095–3166; b) K. H. Shaughnessy, *Chem. Rev.* **2009**, *109*, 643–710; c) F. Joo, *Aqueous Organometallic Catalysis* (Catalysis by Metal Complexes); Kluwer: Dordrecht, Vol. 23, **2001**.
- [2] B. H. Lipshutz, S. Ghorai, A. R. Abela, R. Moser, T. Nishikata, C. Duplais, A. Krasovskiy, *J. Org. Chem.* **2011**, *76*, 4379–4391.
- [3] a) J. Xiang, M. Ipek, V. Suri, W. Massefski, N. Pan, Y. Ge, M. Tam, Y. Xing, J. F. Tobin, X. Xu, S. Tam, *Bioorg. Med. Chem. Lett.* **2005**, *15*, 2865–2869; b) J. Xiang, M. Ipek, V. Suri, M. Tam, Y. Xing, N. Huang, Y. Zhang, J. Tobin, T. S. Mansour, J. McKew, *Bioorg. Med. Chem.* **2007**, *15*, 4396–4405; c) C. Curti, M. Laget, A. O. Carle, A. Gellis, P. Vanelle, *Eur. J. Med. Chem.* **2007**, *42*, 880–884; d) Y. M. Markitanov, V. M. Timoshenko, Y. G. Shermolovich, *J. Sulfur Chem.* **2014**, *35*, 188–236.
- [4] R. E. Swenson, T. J. Sowin, H. Q. Zhang, *J. Org. Chem.* **2002**, *67*, 9182–9185.
- [5] C.-K. Chan, Y.-H. Chen, R.-T. Hsu, M.-Y. Chang, *Tetrahedron* **2017**, *73*, 46–54.
- [6] N.-C. Hsueh, H.-Y. Chen, M.-Y. Chang, *J. Org. Chem.* **2017**, *82*, 13324–13332.
- [7] N.-C. Hsueh, Y.-T. Hsiao, M.-Y. Chang, *Tetrahedron* **2017**, *73*, 4398–4406.
- [8] a) J. L. Marco, I. Fernandez, N. Khair, P. Fernandez, A. Romero, *J. Org. Chem.* **1995**, *60*, 6678–6679; b) J. L. Marco, *J. Org. Chem.* **1997**, *62*, 6575–6581.
- [9] a) H. Yang, R. G. Carter, L. N. Zakharov, *J. Am. Chem. Soc.* **2008**, *130*, 9238–9239; b) X. Tang, L. Huang, Y. Xu, J. Yang, W. Wu, H. Jiang, *Angew. Chem. Int. Ed.* **2014**, *53*, 4205–4208; *Angew. Chem.* **2014**, *126*, 4289–4292.
- [10] D. Nair, P. Pavashe, I. N. N. Namboothiri, *Tetrahedron* **2018**, *74*, 2716–2724.
- [11] a) N. Suryakiran, T. S. Reddy, K. Ashalatha, M. Lakshman, Y. Venkateswarlu, *Tetrahedron Lett.* **2006**, *47*, 3853–3856; b) A. Kumar, M. K. Muthyala, *Tetrahedron Lett.* **2011**, *52*, 5368–5370; c) N. Suryakiran, P. Prabhakar, K. Rajesh, V. Suresh, Y. Venkateswarlu, *J. Mol. Catal. A* **2007**, *270*, 201–204; d) N. Samakkanad, P. Katrun, T. Techajaronjit, S. Hlekhlai, M. Pohmakotr, V. Reutrakul, T. Jaipetch, D. Soorukram, C. Kuhakarn, *Synthesis* **2012**, *44*, 1693–1699; e) D. Kumar, S. Sundaree, V. S. Rao, R. S. Varma, *Tetrahedron Lett.* **2006**, *47*, 4197–4199.
- [12] a) B. M. Trost, D. P. Curran, *Tetrahedron Lett.* **1981**, *22*, 1287–1290; b) A.-L. Fan, S. Cao, Z. Zhang, *J. Heterocycl. Chem.* **1997**, *34*, 1657–1660.
- [13] a) A. K. Singh, R. Chawla, L. D. S. Yadav, *Tetrahedron Lett.* **2014**, *55*, 4742–4746; b) D. Yang, B. Huang, W. Wei, J. Li, G. Lin, Y. Liu, J. Ding, P. Sun, H. Wang, *Green Chem.* **2016**, *18*, 5630–5634; c) T. Niu, J. Cheng, C. Zhuo, D. Jiang, X. Shu, B. Ni, *Tetrahedron Lett.* **2017**, *58*, 3667–3671; d) A. K. Singh, R. Chawla, T. Keshari, V. K. Yadav, L. D. S. Yadav, *Org. Biomol. Chem.* **2014**, *12*, 8550–8554; e) H. Jiang, Y. Cheng, Y. Zhang, S. Yu, *Eur. J. Org. Chem.* **2013**, 5485–5492.
- [14] a) A. K. Singh, R. Chawla, L. D. S. Yadav, *Tetrahedron Lett.* **2014**, *55*, 2845–2848; b) S. Handa, J. C. Fennewald, B. H. Lipshutz, *Angew. Chem. Int. Ed.* **2014**, *53*, 3432–3435; *Angew. Chem.* **2014**, *126*, 3500–3503; c) L. Wang, W. Wei, D. Yang, H. Cui, H. Yue, H. Wang, *Tetrahedron Lett.* **2017**, *58*, 4799–4802; d) Y.-S. Xiong, J. Weng, G. Lu, *Adv. Synth. Catal.* **2018**, *360*, 1611–1616; e) Q. Lu, J. Zhang, G. Zhao, Y. Qi, H. Wang, A. Lei, *J. Am. Chem. Soc.* **2013**, *135*, 11481–11484; f) S. Cai, D. Chen, Y. Xu, W. Weng, L. Li, R. Zhang, M. Huang, *Org. Biomol. Chem.* **2016**, *14*, 4205–4209; g) T. Niu, D. Jiang, B. Ni, *Tetrahedron Lett.* **2017**, *58*, 4299–4303.
- [15] a) Y. Tang, Y. Fan, H. Gao, X. Li, X. Xu, *Tetrahedron Lett.* **2015**, *56*, 5616–5618; b) V. K. Yadav, V. P. Srivastava, L. D. S. Yadav, *Tetrahedron Lett.* **2016**, *57*, 2236–2238.
- [16] D. Kumar, S. Sundaree, V. S. Rao, R. S. Varma, *Tetrahedron Lett.* **2006**, *47*, 4197–4199.
- [17] V. S. Rawat, P. L. M. Reddy, B. Sreedhar, *RSC Adv.* **2014**, *4*, 5165–5168.
- [18] X.-W. Lan, N.-X. Wang, C.-B. Bai, W. Zhang, Y. Xing, J.-L. Wen, Y.-J. Wang, Y.-H. Li, *Sci. Rep.* **2015**, *5*, 18391–18398.
- [19] S. Deng, E. Liang, Y. Wu, X. Tang, *Tetrahedron Lett.* **2018**, *59*, 3955–3957.
- [20] a) Y. Fu, M.-P. Li, C.-Z. Shi, F.-R. Li, Z. Du, C. Huo, *Org. Biomol. Chem.* **2019**, *17*, 10172–10177; b) Y. Fu, Q.-S. Xu, C.-Z. Shi, C.-Q. Xiao, Z. Du, *Adv. Synth. Catal.* **2018**, *360*, 3502–3506; c) Y. Fu, Q.-Z. Li, Q.-S. Xu, H. Hügel, M.-P. Li, Z. Du, *Eur. J. Org. Chem.* **2018**, *2018*, 6966–6970; d) Y. Fu, Q.-S. Xu, Q.-Z. Li, Z. Du, K.-H. Wang, D. Huang, Y. Hu, *Org. Biomol. Chem.* **2017**, *15*, 2841–2845; e) Y. Fu, Y. Su, Q.-S. Xu, Z. Du, Y. Hu, K.-H. Wang, D. Huang, *RSC Adv.* **2017**, *7*, 6018–6022; f) Y. Fu, W. Zhu, X. Zhao, H. Hügel, Z. Wu, Y. Su, Z. Du, D. Huang, Y. Hu, *Org. Biomol. Chem.* **2014**, *12*, 4295–4299; g) Y. Fu, Q.-S. Xu, Q.-Z. Li, M.-P. Li, C.-Z. Shi, Z. Du, *ChemistryOpen* **2019**, *8*, 127–131.
- [21] a) F. O'Hara, R. D. Baxter, A. G. O'Brien, M. R. Collins, J. A. Dixon, Y. Fujiwara, Y. Ishihara, P. S. Baran, *Nat. Protoc.* **2013**, *8*, 1042–1047; b) F. C. Whitmore, F. H. Hamilton, *Org. Synth.* **1922**, *2*, 89–90.
- [22] Organosulfonyl chloride do not react with ketones under our reaction conditions.

Submitted: February 12, 2020

Accepted: February 28, 2020